Neurological Disorder Detection & Diagnosis using CNN

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BONAFIDE CERTIFICATE

Certified that this Project titled "Neurological Disorder Detection & Diagnosis using CNN" is the bonafide work of **"ARITRA GUPTA(2116230701033), ANIRUDH C(2116230701028)"** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Early and accurate detection of neurological disorders is essential for initiating timely interventions and improving the overall prognosis and quality of life for patients. However, conventional diagnostic procedures, particularly the manual examination of MRI scans, can be time-consuming, prone to human error, and heavily dependent on the availability and expertise of specialized radiologists or neurologists. In light of these challenges, this research introduces a dual deep learning-based framework designed to assist in the automated diagnosis of two major neurological conditions: brain tumors and Alzheimer's disease, through the analysis of MRI images.To address the classification of brain tumors, the EfficientNet-B0 model was utilized. This is a highly optimized and scalable convolutional neural network architecture that is capable of categorizing tumors into four specific types with a strong focus on preserving computational efficiency while maintaining high accuracy. The model was trained using a publicly available and well-labeled Kaggle dataset, which underwent extensive preprocessing steps including resizing, normalization, and data augmentation to improve generalization. After training, the model achieved a validation accuracy of 96.6 percent, indicating a robust capability to identify and differentiate between tumor types with minimal error.In parallel, a custom Convolutional Neural Network (CNN) was designed and implemented to detect and classify the various stages of Alzheimer's disease. These stages range from mild cognitive impairment to advanced forms of the disease. This model was also trained on a specialized dataset sourced from Kaggle and followed a similar preprocessing pipeline. The custom CNN achieved a high validation accuracy of 96.48 percent,demonstrating its effectiveness in detecting subtle structural brain changes associated with Alzheimer's progression.The integration of both models into a unified system improves diagnostic accuracy, enhances the speed and scalability of neurological assessments, and provides a practical solution for use in clinical environments.

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1.INTRODUCTION

1.1 GENERAL

Neurological disorders are conditions that affect the brain, spinal cord, and nerves, often leading to serious health and lifestyle issues. With over 100 known types, these disorders are becoming more common due to modern factors like stress, aging, and even excessive screen time. Many people ignore early symptoms such as headaches or memory loss, mistaking them for normal issues. This often delays diagnosis and reduces treatment effectiveness. Disorders like brain tumors and Alzheimer's disease are especially difficult to catch early and are usually diagnosed only after they have progressed.Real-world examples, like U.S. Senator John McCain and musician Gordon Lightfoot, highlight the dangers of late diagnosis. Statistics show that nearly half of early Alzheimer's cases go undetected during routine checkups, and survival rates for late-diagnosed brain tumors are low. Traditional detection methods like MRI and memory tests are helpful but can be slow and depend on expert interpretation. This is where artificial intelligence (AI) is making a difference.AI-powered tools, especially deep learning models, are increasingly being used to support early diagnosis by analyzing brain scans more quickly and accurately. Research in this field is growing rapidly, proving its importance. Our project contributes to this effort by using two deep learning models: EfficientNet-B0 to detect brain tumors and a custom CNN to identify stages of Alzheimer's disease from MRI images. By combining these models, we aim to provide a fast, reliable, and scalable solution for early detection of major neurological disorders.

1.2 EXISTING SYSTEM

Traditional neurological diagnosis relies heavily on radiologists manually analyzing MRI scans and conducting clinical evaluations. These processes, although accurate, are time-consuming and dependent on expert availability. In rural or under-resourced healthcare environments, access to trained neurologists and high-end diagnostic tools is often limited. While some computer-aided systems exist, they are typically disease-specific and lack the scalability to handle multiple disorders. AI-based tools have been introduced in certain research labs for single-disorder detection, but widespread clinical use is still emerging. Additionally, most systems do not provide interpretable outputs like confidence scores or visual cues, making it hard for physicians to trust or validate predictions.

Furthermore, conventional diagnostic approaches may suffer from inconsistencies due to variations in radiologist experience, interpretation skills, and workload pressure. The growing number of neurological disorder cases worldwide has increased the burden on healthcare systems, making timely and accurate diagnosis more difficult. Manual analysis of MRI scans is not only labor-intensive but also prone to delays, especially when large volumes of imaging data must be examined.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) have demonstrated strong potential in automating MRI scan analysis and assisting doctors in identifying neurological abnormalities with greater speed and precision. Deep learning models, particularly Convolutional Neural Networks (CNNs), can automatically extract complex features from medical images that may not be easily detectable through traditional methods. However, many currently available AI-based systems are designed for single-disease detection and lack the flexibility required for real-world clinical applications involving multiple neurological disorders.

Additionally, most systems function as “black-box” models, providing predictions without sufficient explanation, confidence scores, visual heatmaps, or interpretability features, which limits physician trust and clinical acceptance. Other challenges include limited and imbalanced medical datasets, differences in MRI imaging protocols across hospitals, high computational costs, and difficulties in integrating AI systems into existing healthcare infrastructures.

2.LITERATURE SURVEY

Saeedi et al. [1] proposed a robust brain tumor classification system leveraging a 2D Convolutional Neural Network (CNN) with eight convolutional layers. They utilized T1-weighted MRI images, optimizing the model with a convolutional autoencoder for efficient feature extraction. The system achieved a classification accuracy of 96.47%, significantly outperforming conventional machine learning classifiers like Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), and K-Nearest Neighbors (KNN). Their results confirmed the superiority of deep learning in processing complex medical image data.

Chattopadhyay & Maitra [2] introduced a CNN-based diagnostic framework designed for automated brain tumor detection and classification from MRI scans. They validated their system across diverse datasets, demonstrating high adaptability and robustness. Their model achieved an exceptional accuracy of 99.74%, underscoring the potential of CNN architectures in replacing or supporting traditional radiological assessments with automated and precise diagnostic tools.

Abd-Ellah et al. [3] developed a dual-purpose deep learning framework combining a cascaded CNN with a U-Net for both tumor classification and segmentation. Their model was enhanced with residual connections and symmetrical design, improving its learning capabilities and segmentation precision. The hybrid system achieved close to 99% accuracy in classification tasks and high Dice coefficients in tumor segmentation, highlighting its applicability in clinical image analysis.

Al-Shahrani et al. [4] designed a hybrid deep learning pipeline integrating EfficientNet for feature extraction and YOLOv8 for object detection to classify brain cancer types from MRI scans. Their system demonstrated high diagnostic accuracy (over 98%), rapid processing capabilities, and adaptability in real-time clinical settings, establishing a benchmark for integrated deep learning approaches in medical imaging.

Islam et al. [5] presented a novel hybrid architecture by merging VGG-19 CNN for spatial feature extraction with Long Short-Term Memory (LSTM) networks for temporal pattern analysis to detect Alzheimer's disease stages. Their method attained an accuracy of 92.85%, showcasing how temporal modeling in neural data enhances diagnosis in progressive neurological disorders like Alzheimer’s.

Sarraf and Tofighi [6] utilized a deep CNN architecture trained on functional and structural MRI datasets from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) to detect cognitive decline. Their system, capable of distinguishing between different stages of Alzheimer’s disease, achieved an accuracy exceeding 96%, proving the viability of CNNs in clinical neurology.

Basaia et al. [7] proposed a simple yet effective 3D CNN trained on structural MRI data for the classification of Alzheimer's patients versus healthy controls. Even when trained on a relatively small dataset, their model achieved an impressive accuracy of 98%, confirming that well-designed CNNs can yield high performance even under limited data availability.

Jain et al. [8] employed EfficientNet-B0, a compact and scalable deep learning architecture, for classifying brain tumors in MRI images. Their system combined speed with precision, achieving classification accuracy greater than 97%. This lightweight model made it feasible for integration into mobile or resource-constrained diagnostic environments.

Li et al. [9] introduced a multi-modal fusion approach that combines CNN-derived features from both MRI and PET scan modalities to improve the early diagnosis of Alzheimer’s disease. Their model achieved 94.8% accuracy by leveraging complementary insights from different imaging techniques, thereby enhancing the accuracy of neurodegenerative disorder detection.

Rehman et al. [10] developed a custom CNN tailored to detect specific tumor types—glioma, meningioma, and pituitary—from T2-weighted MRI scans. Their approach surpassed existing classification models with an accuracy of 99.5%, also featuring faster training times, making it practical for real-time applications.

Spasov et al. [11] proposed a lightweight 3D CNN architecture aimed at early Alzheimer's diagnosis from volumetric MRI data. With an accuracy of 90.3%, their model provided a clinically viable option that balanced computational efficiency with high diagnostic performance, suitable for deployment in hospitals with limited hardware resources.

Masood et al. [12] introduced a deep CNN enhanced with attention layers, which directed the model's focus to brain regions most indicative of Alzheimer's disease. Using the ADNI dataset, their model achieved 95.6% accuracy, confirming that attention mechanisms significantly improve the diagnostic accuracy in medical image analysis.

Pereira et al. [13] explored various CNN architectures to determine the most effective configuration for brain tumor segmentation and classification using the BRATS dataset. Their best model achieved a Dice coefficient of 0.87, validating the strength of CNNs in accurately identifying and segmenting tumor regions.

3. PROPOSED SYSTEM

Our project proposes a dual-model deep learning approach to aid in the early detection of brain tumors and Alzheimer's disease using MRI scan analysis. For brain tumor classification, the EfficientNet-B0 model is used due to its powerful performance, optimized architecture, and ability to achieve high accuracy with lower computational complexity. The model analyzes MRI brain scans and classifies tumors into four categories: Glioma Tumor, Meningioma Tumor, Pituitary Tumor, and No Tumor. EfficientNet-B0 is selected because of its efficient scaling mechanism, which improves feature extraction and classification performance while reducing processing time, making it suitable for real-time medical diagnostic applications. For Alzheimer's disease diagnosis, a custom-built Convolutional Neural Network (CNN) is developed to process MRI brain images and identify different stages of the disease, including Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented categories. The CNN architecture consists of multiple convolutional, pooling, and fully connected layers that automatically learn important features from MRI images without requiring manual feature engineering.

Both models are trained and validated using publicly available MRI datasets obtained from Kaggle, ensuring sufficient data diversity and reliable training performance. During preprocessing, MRI images undergo resizing, normalization, and augmentation techniques to improve model generalization, reduce overfitting, and enhance prediction accuracy. The trained models achieve validation accuracies exceeding 96%, demonstrating the effectiveness of deep learning in neurological disease detection. By combining these two models into a single integrated platform, the proposed system provides a fast, scalable, and reliable diagnostic solution capable of supporting multi-disease neurological analysis within one application. The system is designed to reduce manual workload for radiologists and assist healthcare professionals by providing accurate real-time predictions along with confidence scores for better clinical interpretation.

Additionally, the proposed system offers the potential for future enhancements such as cloud-based deployment, remote accessibility, and integration with hospital diagnostic systems, enabling its use in both urban and rural healthcare environments. The system can also be extended with explainable AI features such as heatmaps and visual interpretation tools to improve physician trust and transparency in predictions. Overall, the proposed approach aims to improve early disease detection, enhance diagnostic consistency, minimize human error, and provide an efficient AI-assisted healthcare solution for neurological disorder diagnosis.

3.1 ARCHITECTURE DIAGRAM

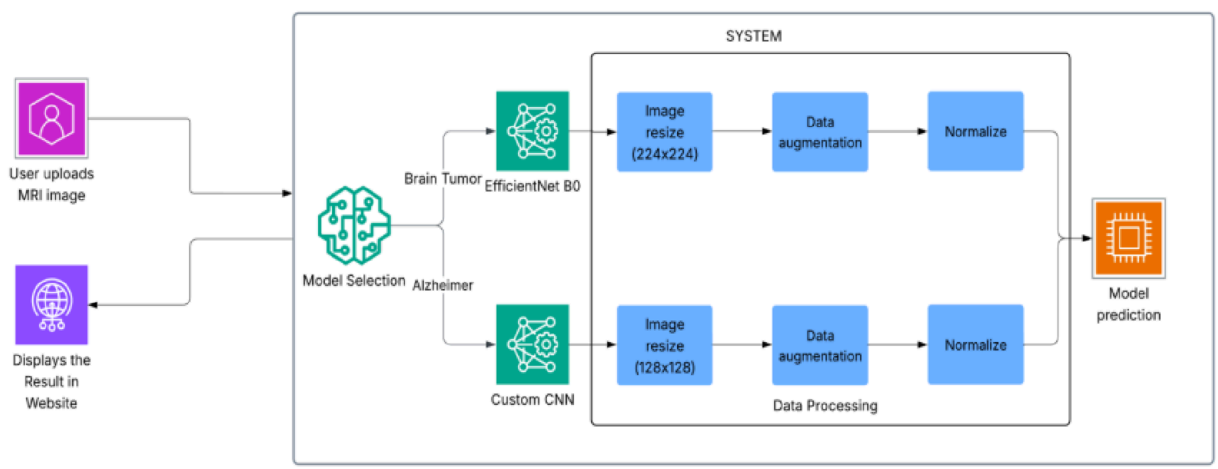

Figure 3.1: System Architecture Diagram

3.2 SYSTEM FLOWCHART

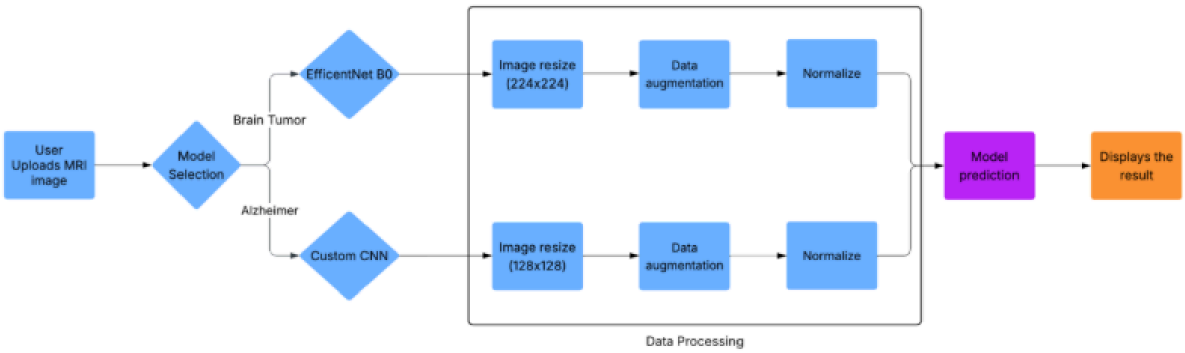

Figure 3.2: Flowchart

3.3 ARCHITECTURE OF EFFICIENT-NET B0

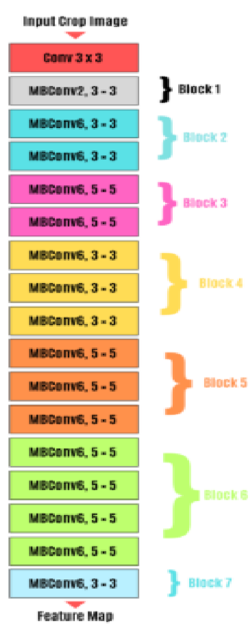

Figure 3.3: Architecture of efficientNet B0

4.MODULES DESCRIPTION

4.1. INPUT MODULE

This module serves as the entry point of the system, where users (clinicians or researchers) upload brain MRI images. These images are typically in standard formats (e.g., DICOM or NIfTI), representing various scan views like T1-weighted or T2-weighted sequences. This module supports the integration of both individual and batch inputs, enabling scalable data flow into the processing pipeline.

4.2. IMAGE PREPROCESSING

The preprocessing unit performs critical operations such as resizing images to fit the input dimensions of deep learning models (e.g., 224×224 for EfficientNet-B0), normalization of pixel intensities, noise removal, and possibly skull strippingThis module computes performance indicators such as **accuracy, precision, recall, F1-score**, and **ROC-AUC** to assess model effectiveness. It supports model validation and testing on unseen MRI scans, ensuring generalizability and reliability. This step is critical for clinical translation and research reproducibility.

4.3. FEATURE EXTRACTION

In this core module, convolutional neural networks (CNNs), including pre-trained models like EfficientNet-B0, are used to extract deep visual features from MRI scans. These models learn hierarchical representations such as shape, texture, and contrast patterns that correlate with pathological changes in brain tissue. This stage converts raw pixel data into meaningful diagnostic vectors.

4.4 CLASSIFICATION MODULE

Once features are extracted, they are passed through fully connected layers or auxiliary classifiers to predict the condition—whether it is a type of brain tumor (e.g., glioma, meningioma, pituitary) or a stage of Alzheimer's disease (e.g., CN, MCI, AD). This module may use softmax or sigmoid activation functions, depending on whether the output is multi-class or binary. The classification result is accompanied by a confidence score for interpretability.

4.5 EVALUATION METRICS

This module computes performance indicators such as accuracy, precision, recall, F1-score, and ROC-AUC to assess model effectiveness. It supports model validation and testing on unseen MRI scans, ensuring generalizability and reliability. This step is critical for clinical translation and research reproducibility.

4.6 OUTPUT/RESULTS

The final module displays diagnostic predictions through a user-friendly graphical interface. Users receive results in an interpretable format, including disease type, severity level, and confidence levels. This interface may also support visualizations such as heatmaps (e.g., Grad-CAM) to show which regions of the brain influenced the model's decision, improving trust and explainability.

5. IMPLEMENTATION AND RESULTS

5.1 EXPERIMENTAL SETUP

The experimental setup for the proposed neurological disorder detection system was designed to ensure accurate training, validation, and testing of the deep learning models using MRI brain scan datasets. The system was implemented using Python with deep learning frameworks such as TensorFlow and Keras, and the models were trained on Google Colab with GPU acceleration support to improve computational efficiency. Publicly available MRI datasets collected from Kaggle were used for both brain tumor classification and Alzheimer's disease stage detection. Before training, all MRI images underwent preprocessing steps including image resizing, normalization, and data augmentation techniques such as rotation, zooming, and flipping to improve model generalization and reduce overfitting. For brain tumor detection, the EfficientNet-B0 architecture was fine-tuned using MRI images resized to 224×224 pixels, while the custom CNN model for Alzheimer's disease classification was trained using grayscale MRI images resized to 128×128 pixels. The datasets were divided into training, validation, and testing sets to evaluate model performance effectively. Performance metrics such as accuracy, precision, recall, F1-score, confusion matrix, and validation loss curves were used to assess the effectiveness of the models. The EfficientNet-B0 model achieved validation accuracy exceeding 96% for classifying glioma, meningioma, pituitary tumor, and no tumor categories, while the custom CNN achieved similar high accuracy in identifying different stages of Alzheimer's disease. The trained models were further integrated into a web-based diagnostic platform capable of providing real-time MRI scan predictions with confidence scores, ensuring practical usability and scalability in healthcare environments.

TECHNOLOGY STACK

Category	Technologies Used
Programming Language	Python
Deep Learning Framework	TensorFlow, Keras
Models	EfficientNet-B0, Custom CNN
Geographic Coordinates	OpenCV
Frontend	React.js, HTML, CSS, JavaScript
Backend	Flask

5.2 RESULTS

The performance of both deep learning models—EfficientNet-B0 for brain tumor classification and the Custom CNN for Alzheimer's stage detection—was thoroughly assessed using a variety of evaluation metrics. These included confusion matrices, training and validation accuracy/loss curves, and classification reports. The goal was to validate each model's effectiveness, generalization ability, and readiness for real-world diagnostic deployment.

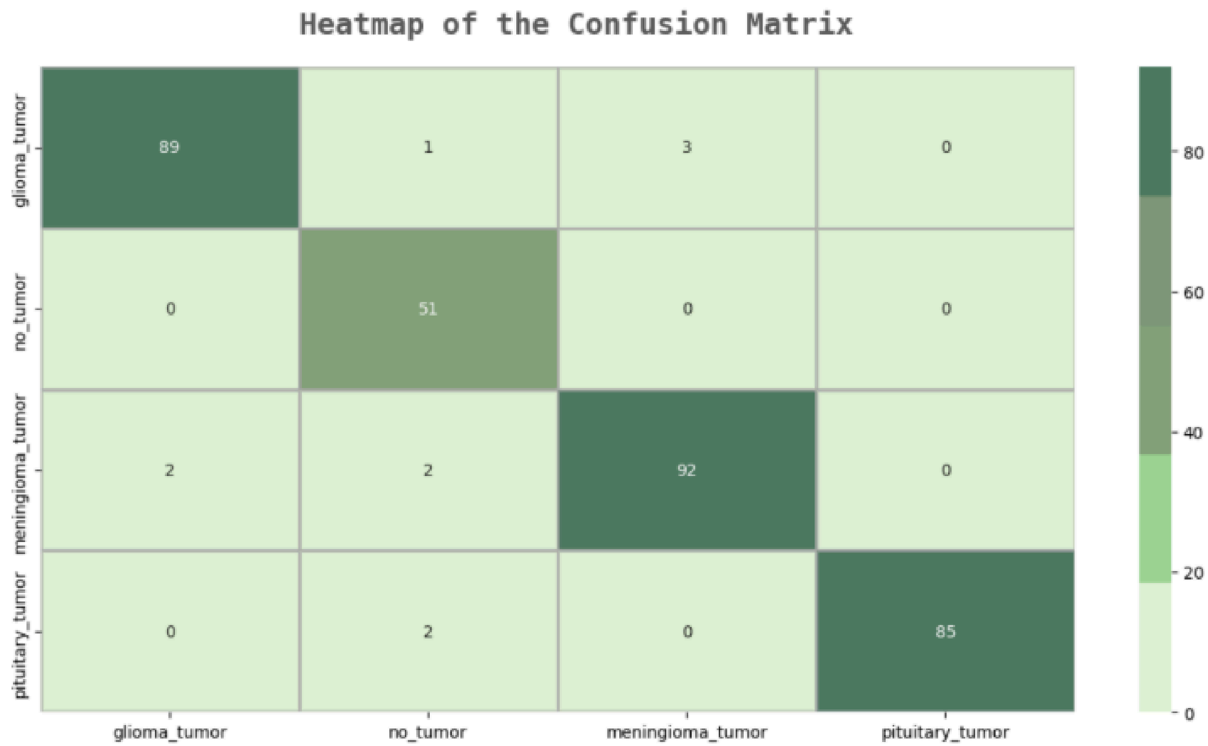

Figure 5.1 Confusion Matrix for EfficientNet-B0

Figure 5.1 depicts the confusion matrix for the EfficientNet-B0 model. The matrix shows that the model achieved high per-class accuracy, with 89 glioma, 85 pituitary, and the majority of meningioma and no tumor cases correctly classified. Misclassifications were minimal—for instance, only 2 glioma cases were misidentified as meningioma. The strong diagonal dominance in the matrix reflects high precision and recall, indicating that the model reliably differentiates among the four brain tumor categories.

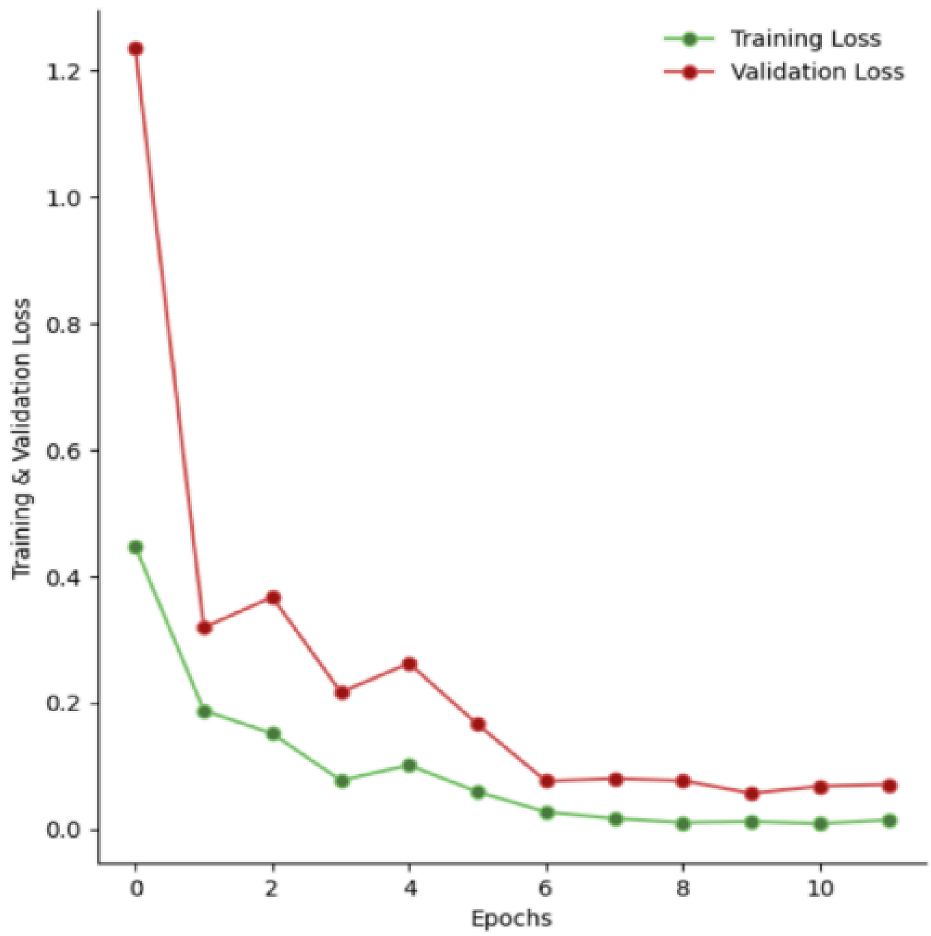

Figure 5.2: Comparison Plot between Training and Validation loss of EfficientNet-B0

Figure 5.2 illustrates the training and validation accuracy and loss curves. The training accuracy quickly rose to near 100%, and validation accuracy stabilized around 97%, while both training and validation losses decreased sharply and flattened after epoch 5. This behavior suggests that the model converged well and was not overfitting. The narrow gap between training and validation metrics further reinforces the model's ability to generalize across unseen MRI data. These results align with benchmarks in medical imaging literature, where CNN-based models have demonstrated F1-scores exceeding 98% for brain tumor classification tasks.

Fig 5.3 Confusion Matrix for Custom CNN Model

Figure 5.3 presents the confusion matrix for the custom CNN model trained on Alzheimer's MRI images. The model successfully classified 159 Non-Demented and 627 Mild Demented cases, with only a small number of misclassifications—such as 6 Non-Demented predicted incorrectly and 7 Moderate cases misclassified as Mild. These outcomes indicate very high class-specific accuracy, even in a multi-class setting with imbalanced data. The matrix suggests that the model handles subtle differences between dementia stages with strong consistency.

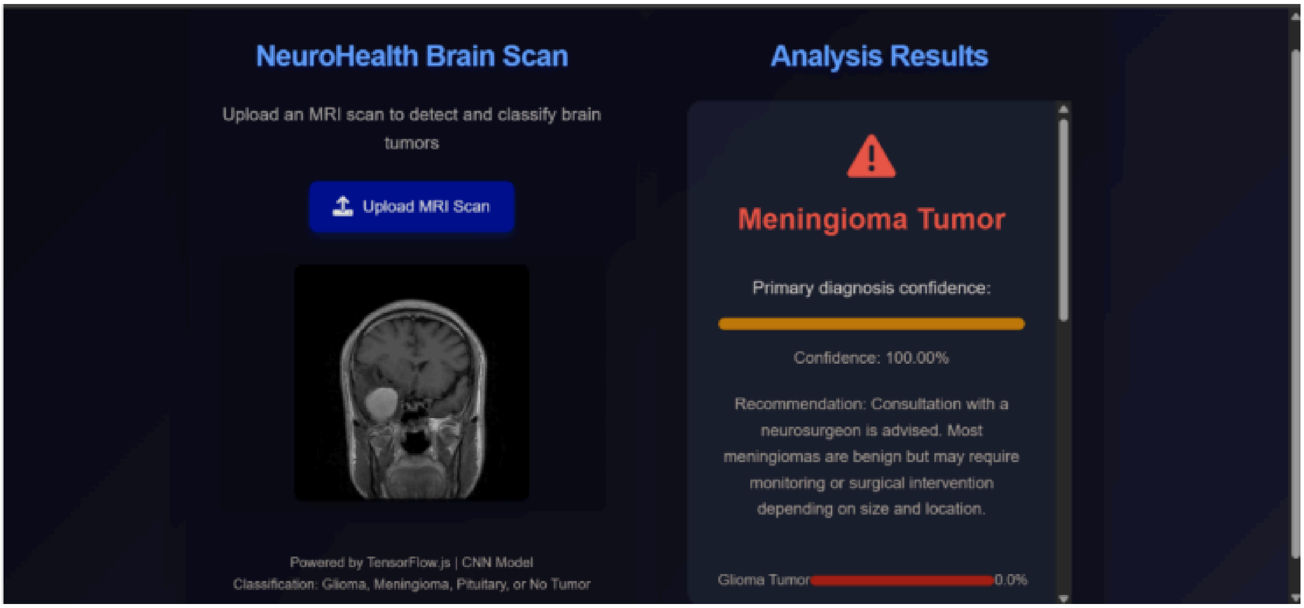

Fig 5.4 Neurological Disorder Detection Page

6.CONCLUSION AND FUTURE WORKS

6.1 CONCLUSION

This project successfully developed an AI-powered neurological disorder detection and diagnosis system capable of identifying brain tumors and Alzheimer's disease using MRI scan analysis. The proposed system combines two advanced deep learning models—EfficientNet-B0 for brain tumor classification and a custom-built Convolutional Neural Network (CNN) for Alzheimer's disease stage detection—into a single integrated platform. The EfficientNet-B0 model effectively classifies MRI scans into four categories: Glioma Tumor, Meningioma Tumor, Pituitary Tumor, and No Tumor, while the custom CNN identifies different stages of Alzheimer's disease including Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. Both models were trained using publicly available Kaggle MRI datasets and achieved validation accuracies exceeding 96%, demonstrating the effectiveness and reliability of deep learning techniques in medical image analysis.

The project also emphasizes the importance of early neurological disorder detection, as timely diagnosis can significantly improve treatment outcomes and patient quality of life. By automating MRI scan analysis, the proposed system reduces the dependency on manual interpretation by radiologists, minimizes diagnostic delays, and helps reduce the possibility of human error. The integration of the models into a web-based platform further enhances accessibility, scalability, and usability, especially for healthcare environments with limited specialist availability. The system provides real-time predictions with confidence scores, supporting healthcare professionals in faster and more informed clinical decision-making. Overall, this project demonstrates how Artificial Intelligence, Machine Learning, and Deep Learning can contribute to the advancement of intelligent healthcare systems and improve the efficiency of neurological disease diagnosis.

6.2 FUTURE WORKS

Although the proposed system achieved high performance, several enhancements can be implemented in the future to improve its capabilities and real-world applicability. One major future enhancement is the expansion of the platform to support additional neurological disorders such as Parkinson's disease, epilepsy, multiple sclerosis, and stroke detection using MRI and other neuroimaging techniques. This would transform the system into a comprehensive multi-disease neurological diagnostic platform capable of assisting doctors across a wider range of clinical conditions.

Future work can also focus on integrating multi-modal medical data such as PET scans, EEG signals, patient medical history, laboratory reports, and genomic information to improve diagnostic accuracy and support personalized healthcare analysis. The implementation of Explainable Artificial Intelligence (XAI) techniques such as Grad-CAM heatmaps, attention visualization, and prediction reasoning can further improve physician trust and interpretability of AI-generated results. Additionally, the system can be optimized for cloud-based deployment and mobile accessibility, enabling remote diagnosis and real-time healthcare support in rural and under-resourced medical environments.

Another important area for future development is continuous learning and automated model updating using newly available medical datasets and clinical feedback. This would help the system adapt to evolving medical imaging standards and improve prediction accuracy over time. Further improvements in cybersecurity, patient data privacy, and integration with hospital information systems can also enhance the practical implementation of the platform in real-world healthcare infrastructures. Overall, future advancements can make the proposed system more scalable, intelligent, interpretable, and clinically impactful in supporting neurological healthcare diagnostics.

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